HSM: Workload-Similarity Gating for Index-Maintenance Decisions, with Formal Bounds

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Abstract—Existing index advisors (Dexter, Supabase `index_advisor`, BALANCE, Indexer++) answer *which* indexes to build but invoke their $\Omega(N \log N)$ search at every window or on a fixed schedule. We address the complementary question—*when* should the advisor be invoked?—with HSM (Hierarchical Similarity Measurement), a training-free, DBMS-agnostic decision layer that gates advisor invocation in $\Theta(N_{\text{pts}})$ time. HSM quantifies workload similarity along five dimensions (template rank, volume, type, attribute overlap, DWT-based temporal pattern) and exposes a single calibrated threshold $\theta^*(N, Q)$. Six theorems and four lemmas establish metric soundness, complexity, economic gating optimality, detector-quality bounds, a deployment speedup certificate, and a convex-optimal window size. On PostgreSQL 16 with TPC-H (SF 0.2–3.0), HSM attains Youden $J \geq 0.96$ at every scale on the phase-labelled retain-vs-rebuild detection protocol (within-phase vs. cross-phase window pairs, $m_w = m_c = 160$, §V.A14), and reduces advisor invocations by 80.8% versus always-on and 42.5% versus periodic ($K = 3$), at a kernel cost of ≈ 1 ms per decision. End-to-end deployment with two production PostgreSQL advisors (Dexter, Supabase `index_advisor`) demonstrates 66–71% wall-clock savings versus always-on with 100% trigger precision. Four external workloads (SDSS, JOB/IMDB, pgbench, burst) and cross-engine validation on MongoDB 7 confirm the kernel transfers unmodified; only θ requires per-engine recalibration—no kernel retraining.

Index Terms—Decision layer, index-maintenance timing, proactive index management, workload similarity, hierarchical similarity measurement, discrete wavelet transform, self-tuning databases.

1 Introduction

Database index performance is sensitive to workload characteristics. When the distribution of query types, access frequency, or temporal patterns changes substantially, an index configuration that was optimal for a previous workload window may become inefficient for the current one. Proactive index management—deciding whether to retain an existing index or rebuild it based on workload evolution—requires a reliable mechanism to determine how similar the current workload window is to the reference window under which the index was last calibrated.

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Prior work [1], [2] validated the HSM concept, demonstrating 77% classification accuracy and a 4,623× latency improvement over unoptimised baselines using a

coarse interval-overlap metric $S_i = |T_{\text{overlap}}|/|T_{\text{total}}|$. That metric cannot distinguish burst events, periodic sub-patterns, or multi-scale frequency structure, and those studies provided no formal theoretical analysis.

We introduce the HSM framework, making five contributions. **C1** A five-dimensional similarity measure with formal metric properties, capturing burst events, periodic oscillations (detected via angular cosine for phase-alignment), and long-term trends simultaneously. **C2** A theoretical core of six theorems and four lemmas covering wavelet pattern preservation, minimal-sufficient dimensions, economic gating optimality, stratified-Hoeffding detector-quality bounds, a deployment speedup certificate, and a convex-optimal window size (Section IV). **C3** Access-attribute independence: HSM incurs no reconstruction cost when the accessed column set changes (Theorem 3, part (ii); supplementary §VII). **C4** A closed-form threshold $\theta^*(N, Q)$ validated across SF 0.2–3 (Youden $J \geq 0.96$ at every scale under the phase-labelled detection protocol) and generalising across TPC-H, burst, and cross-engine workloads. **C5** Cross-engine validation on MongoDB 7, confirming that the HSM kernel is DBMS-agnostic and requires only per-engine θ recalibration—no kernel retraining.

2 Related Work

2.1 Workload Characterisation and Similarity Methods

Early workload compression [3] and automated index selection [4] treat workloads as static query sets via predicate overlap or template matching. Recent advisors take richer views but address *what* to index rather than *when*: BALANCE [5] uses self-supervised contrastive learning over index-selection instances; MFIX [6] applies multi-fidelity Bayesian optimisation; Indexer++ [7] detects binary trend shifts with BERT embeddings and deep RL. All assume the advisor is always invoked, and most rely on GPUs and labelled corpora that may be unavailable cross-engine. ShapeShifter [8] and the benchmark of [9] use similarity only as a binary trigger. Prior HSM work [1], [2] validated the concept empirically; the present paper adds formal proofs and scale-sensitivity analysis.

Gap 2.1: no prior method jointly provides a continuous multi-dimensional similarity score with formal metric properties, a closed-form decision threshold, and a training-free deployment model.

Deployed Index Advisors. Alongside academic prototypes, two open-source PostgreSQL advisors define the practical deployment state of automatic index recommendation: *Dexter* [10], which scans query logs and applies what-if analysis via HypoPG [11][1] to recommend B-tree indexes; and Supabase `index_advisor` [12], a PostgreSQL extension that drives the same HypoPG mechanism from an SQL interface. Both are widely deployed (Heroku/Instacart/Supabase production use), actively maintained through 2024, and have been adopted as baseline advisors in several recent industrial and academic studies. Open-source commercial derivatives include *pganalyze* Index Advisor and Huawei's *DBMind* [13]; the latter applies a reinforcement-learning policy in openGauss (a PostgreSQL fork) and therefore lies outside the vanilla-PostgreSQL protocol used in our experiments. Neither Dexter nor Supabase `index_advisor` addresses *when* to invoke advisor analysis—both assume the caller decides invocation externally—so HSM composes with them orthogonally by supplying the missing timing layer (Table 2). Both advisors are deployed end-to-end with HSM as the gating layer in §5.5 (A13d), with measured wall-clock, advisor cost, and trigger precision reported in Table 3.

2.2 Signal Processing for Time-Series Similarity

DWT [14] decomposes signals into multi-resolution components; Chan & Fu [15] applied this to time-series indexing. SAX [16] converts normalised series to discrete symbols with bounded error; a 2025 survey [17] reviews variants, including Quantile SAX [18] for non-Gaussian series. DTW [19] measures elastic similarity; FastDTW [20] approximates DTW in linear time, enabling $O(N)$ workload-window comparison. The three are complementary—multi-resolution decomposition (DWT), bounded-error symbolisation (SAX), and elastic alignment (FastDTW)—motivating the three-stage S_P pipeline (Section III).

Change-point detection (CPD). Classical sequential CPD includes CUSUM [21] and Bayesian Online Change-Point Detection [22], which optimise a detection-delay/false-alarm trade-off by scanning a *stream* position-by-position. HSM differs in protocol: it is a *window-pair similarity classifier* that decides retain-vs-rebuild on aligned windows, not on stream positions (Theorem 4 reformulates the gate as a binary detector under this protocol). The two families are complementary—streaming CPD can supply windows to HSM dynamically, while HSM supplies a DBMS-agnostic composite similarity that single-signal CPD does not provide—and are therefore not substitutable baselines.

Theoretical foundations. Metric space theory [23] formalises similarity properties. Information-theoretic bounds for symbolic representations draw on Cover & Thomas [24]. Wavelet approximation bounds are established in Mallat [25]. These classical results provide the mathematical scaffolding for Section IV.

1. HypoPG creates hypothetical indexes whose presence can be tested through the PostgreSQL query planner without the $\Theta(N \log N)$ cost of physical construction.

Gap 2.2: no prior work provides formal guarantees for multi-scale workload similarity—neither metric properties nor parameter optimality bounds.

2.3 Database Optimisation and Access-Attribute Dependence

Self-tuning systems (AI-DBA [26], CDBTune [27], IA[2] [28], GPTuner [29]) all depend on workload characterisation as an upstream step; none provides a principled multi-dimensional similarity score with formal properties.

All physical index methods share *access-attribute dependence*: the structure is built for a fixed column set, and changing that set requires full reconstruction ($\Theta(N \log N)$). This holds from classical B-Trees [30] and Database Cracking [31], through first-generation learned indexes [32] (ALEX [33], PGM-Index [34], XIndex [35]), to multi-dimensional extensions (Flood [36], Tsunami [37], WaZI [38]). A formal comparison appears in the Supplementary (Structural Comparison with Prior Methods).

Gap 2.3: no existing framework (B-Trees, cracking, learned indexes, WaZI) formally characterises access-attribute independence or provides multi-dimensional temporal similarity with cost guarantees.

2.4 NoSQL and Document Store Indexing

Cattell's taxonomy [39] established the SQL/NoSQL divide as a data-model and consistency tradeoff, but did not address workload similarity measurement. On the schema side, NoSE [40] formulates NoSQL physical design as an integer program over a conceptual data model; however, it targets Cassandra's wide-column model and does not consider temporal workload drift. Klettke et al. [41] address schema evolution and data migration in document stores at scale, highlighting that agile NoSQL development produces frequent schema versions—a setting where proactive drift detection is especially valuable. De Espona Pernas et al. [42] propose query-only automatic index suggestion for MongoDB aggregation pipelines, the closest NoSQL-specific index advisor to date; their method does not model temporal windows or workload similarity. On the SQL side, industrial index managers such as AIM [43] and workload-reduction techniques like Wred [44] remain relational-only. Stonebraker and Pavlo's recent retrospective [45] observes that relational and document models increasingly coexist in modern deployments, yet no cross-engine workload similarity framework has been proposed.

Gap 2.4: no existing framework provides a DBMS-agnostic workload similarity measure validated across SQL and document-store engines with formal metric guarantees.

3 The HSM Framework

HSM compares two workload windows W_A, W_B, each of duration T containing N_{pts} time-stamped query events, in four phases: preprocessing, five-dimensional similarity computation, score aggregation, and index decision.

3.1 Workload Preprocessing

Each window W is reduced to: a k-element query-template frequency vector $\mathbf{f} \in \mathbb{R}^k$ (one entry per distinct template, extracted by the DBMS-specific extractor; Section III-B); a QPS time series $q(t)$ at 1-second resolution giving N_{pts} points[2]; relation set $\mathcal{T}$ and column set $\mathcal{C}$ forming the access-attribute schema $A(W) = (\mathcal{T}, \mathcal{C})$; and total volume $\text{QPS} = \sum_i f_i / T$.

3.2 Five Similarity Dimensions

Ratio Similarity S_R measures the relative-frequency distribution of query templates via Spearman rank correlation on the normalised query-frequency rank vectors $\mathbf{r}$ (one entry per distinct template, derived from $\mathbf{f}$):

$$S_R(W_A, W_B) = 1 - \frac{\arccos\big(\rho_s(\mathbf{r}_A, \mathbf{r}_B)\big)}{\pi}, \tag{1}$$

where $\rho_s(\cdot)$ is the Spearman rank correlation. Since $\rho_s \in [-1, 1]$, we have $\arccos(\rho_s) \in [0, \pi]$, so $S_R \in [0, 1]$. Importantly, $d_R = 1 - S_R = \arccos(\rho_s)/\pi$ is the normalised *correlation angle* and satisfies the triangle inequality analytically [23]. $S_R = 1$ iff both windows share identical query-frequency rank order. **Volume Similarity** S_V uses log-scale to handle order-of-magnitude differences:

$$S_V(W_A, W_B) = \exp(-|\ln Q_A - \ln Q_B|) = \frac{\min(Q_A, Q_B)}{\max(Q_A, Q_B)}. \tag{2}$$

Type Similarity S_T compares query-template distributions via angular distance on unit-normalised frequency vectors. Let $\mathcal{Q} = \{q_1, \ldots, q_k\}$ be the set of k distinct operation categories observed in the monitoring period (e.g. SQL query templates {Q1, Q3, ...} for relational DBMS; pipeline categories {`find`, `aggregate`, `insert`, `update`, `delete`} for document DBMS). Define $\mathbf{v} \in \mathbb{R}^k$ as the normalised frequency count over $\mathcal{Q}$ within window W:

$$S_T(W_A, W_B) = 1 - \frac{2}{\pi} \arccos\big(\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B\big), \tag{3}$$

where $\hat{\mathbf{v}}$ denotes the ℓ_2-normalised unit vector. Frequency counts are non-negative, so $\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B \in [0, 1]$, giving $S_T \in [0, 1]$. The dissimilarity $d_T = (2/\pi) \arccos(\hat{\mathbf{v}}_A \cdot \hat{\mathbf{v}}_B)$ is the *angular distance* (geodesic) on $\mathbb{S}^{k-1}$, satisfying the triangle inequality for any $k \geq 1$ [23]. This DBMS-agnostic formulation allows $\mathcal{Q}$ to be defined by a *DBMS-specific extractor* (Section III-B), making S_T applicable across heterogeneous database paradigms.

Access Similarity S_A measures schema-level overlap as an equal-weight average of two Jaccard scores over table-level and column-level access sets:

$$S_A = \tfrac{1}{2} J(\mathcal{T}_A, \mathcal{T}_B) + \tfrac{1}{2} J(\mathcal{C}_A, \mathcal{C}_B), \quad J(X, Y) = \frac{|X \cap Y|}{|X \cup Y|}. \tag{4}$$

Equal weighting ensures that structural access and fine-grained attribute access carry equal diagnostic weight.

Pattern Similarity S_P captures multi-scale temporal behaviour through a three-stage pipeline. *Stage 1 (DWT).* Daubechies-4 (db4) at level $L = 3$ decomposes $q(t)$ into approximation coefficients $\mathbf{cA}_3$ and detail coefficients $\mathbf{cD}_3$, $\mathbf{cD}_2$, $\mathbf{cD}_1$ at three scales. *Stage 2 (SAX).* Each sub-band coefficient vector is encoded via SAX with alphabet size $\alpha = 4$. *Stage 3 (FastDTW).* FastDTW with Sakoe–Chiba radius $r = 3$ computes the alignment distance; the per-band score is normalised to $[0, 1]$: $\text{band_score} = 1 - \text{dist}/(\text{len}(s) \cdot (\alpha - 1))$.

$$S_P = w_3^A \text{sc}(\mathbf{cA}_3) + w_3^D \text{sc}(\mathbf{cD}_3) + w_2^D \text{sc}(\mathbf{cD}_2) + w_1^D \text{sc}(\mathbf{cD}_1), \tag{5}$$

with $w_3^A = 0.40$, $w_3^D = w_2^D = w_1^D = 0.20$ (sum = 1), where $\text{sc}(\cdot)$ denotes the per-band normalised score. The level-3 approximation band $\mathbf{cA}_3$ receives greater weight (0.40) to capture long-range trend; the three detail bands carry equal weight (0.20 each) to detect burst and periodic events at multiple scales.[3]

3.3 DBMS-Specific Feature Extraction

The five HSM dimensions define a *universal similarity kernel* whose inputs are DBMS-agnostic WorkloadWindow objects. A lightweight *extractor* bridges the gap between raw DBMS traces and these objects, encapsulating all DBMS-specific semantics:

Relational extractor (PostgreSQL/MySQL). $\mathcal{Q}$ = set of SQL query templates identified by the query planner's normalised statement text. $\mathcal{A}$ = {table.column} sets derived from the query access map. Volume Q = query count per window.

Document extractor (MongoDB). $\mathcal{Q}$ = operation categories {`find`, `aggregate`, `insert`, `update`, `delete`}, further resolved to pipeline-stage signatures for aggregation queries. $\mathcal{A}$ = {`collection.field_path`} accessed per operation (respecting nested-document paths). Volume Q = operation count per window.

The similarity kernel (formulas for S_R–S_P) is identical for both extractors; only the vocabulary $\mathcal{Q}$ and attribute set $\mathcal{A}$ change. DBMS-specific default weights are set to balanced priors and validated by sensitivity analysis on representative traces (Section 3.4, Experiment A10b).

3.4 HSM Score Aggregation

$$S_{\text{HSM}}(W_A, W_B) = \sum_{i=1}^{5} w_i S_i(W_A, W_B), \tag{6}$$

with $\sum_i w_i = 1$, $w_i \geq 0.05$ (all dimensions necessary, Theorem 2). Default weights are chosen as balanced priors, then validated by sensitivity analysis (A10b): DR changes by at most 5.5% under ± 0.05 single-weight perturbation. PostgreSQL defaults: $w_R = 0.25$, $w_V = w_T = w_A = 0.20$, $w_P = 0.15$. Table 1 summarises the five dimensions.

2. The implementation normalises $q(t)$ by $\max_t q(t)$ before the wavelet decomposition (`hsm_v2_kernel.py`: `sp_v2`). This makes the bound constant $M = 1$ in Theorem 1, so the cited tight bound $\varepsilon \leq 0.071$ at $L = 3$, $N_{\text{pts}} = 100$ holds without further scaling.

3. To enforce the metric properties of Lemma 1 the implementation (`hsm_v2_kernel.py`: `_band_score`) (i) returns 1.0 when the two SAX encodings are byte-identical (P2 self-distance) and (ii) symmetrises the FastDTW alignment by averaging $\frac{1}{2}[\text{fastdtw}(s_a, s_b) + \text{fastdtw}(s_b, s_a)]$ (P3 symmetry). FastDTW is an approximation of full DTW, so without these two corrections P2 and P3 hold only up to the FastDTW radius; the implementation therefore satisfies Lemma 1 more tightly than the reference statement.

TABLE 1
Summary of HSM similarity dimensions. Each dimension maps a pair of workload windows to $[0, 1]$; the composite S_{HSM} is a convex combination (Eq. 6). Cost is per window-pair evaluation.

Dim.	Formula	Range	Cost	Captures
S_R	Spearman+arccos	$[0, 1]$	$O(N_{\text{pts}})$	Query-type freq. ranks
S_V	Log-exp	$(0, 1]$	$O(N_{\text{pts}})$	Query volume
S_T	Angular dist.	$[0, 1]$	$O(k)$	Query-template dist.
S_A	Avg. Jaccard	$[0, 1]$	$O(\|A\|)$	Attribute overlap
S_P	DWT+SAX+DTW	$[0, 1]$	$O(N_{\text{pts}})$	Temporal pattern
S_{HSM}	Convex sum	$[0, 1]$	$O(N_{\text{pts}})$	All five dims

3.5 Index Decision Rule

Definition 1 (Decision Threshold θ)**.** *The threshold $\theta \in (0, 1)$ is the minimum HSM score below which the expected cost of retaining the current index exceeds the expected cost of rebuilding it. Given $h = S_{\text{HSM}}(W_{\text{cur}}, W_{\text{ref}})$: if $h \geq \theta^*(N, Q)$ then* RETAIN*; if $h < \theta^*(N, Q)$ then* REBUILD*. The closed-form derivation of θ^* is in Theorem 3.*

3.6 Algorithm 2: HSM Index Diagnosis

When the gate fires ($h < \theta^*$), the operator needs to know *which* dimension is responsible for the regime shift in order to target a single index repair instead of a full rebuild. Algorithm 2 implements that diagnosis as a one-shot dimension-wise contribution analysis that is $O(1)$ on top of an HSM evaluation.

Algorithm 2 (HSM Index Diagnosis).
Input: per-dimension scores $\{S_d\}_{d \in \mathcal{D}}$ with $\mathcal{D} = \{R, V, T, A, P\}$ and weights $\{w_d\}$ from W_0.
Output: blamed dimension d^* and contribution vector $\mathbf{c}$.

1) For each $d \in \mathcal{D}$, compute the shortfall contribution $c_d = w_d\,(1 - S_d)$.
2) Return $d^* = \arg\max_d c_d$ and the full vector $\mathbf{c} = (c_R, c_V, c_T, c_A, c_P)$.
3) *Counterfactual repair check:* compute $\Delta S_{\text{HSM}} = w_{d^*}(1 - S_{d^*})$, the closed-form gain to HSM if dimension d^* were perfectly repaired.

The decomposition is exact because $1 - S_{\text{HSM}} = \sum_d w_d(1 - S_d)$, so $c_d/(1 - S_{\text{HSM}})$ is the share of the gap attributable to dimension d. Algorithm 2 thus turns a scalar gate decision into an attributable diagnosis without any extra DBMS state and without any tuning.

Robustness model. In deployment the operator measures noisy estimates $\hat{S}_d = \text{clip}(S_d + \varepsilon_d,\, 0,\, 1)$ with $\varepsilon_d \sim \mathcal{N}(0, \sigma^2)$ i.i.d. across dimensions. Algorithm 2 then chooses $\hat{d}^* = \arg\max_d w_d(1 - \hat{S}_d)$, which can disagree with the clean argmax when two dimensions have similar contributions. The operator's repair, however, acts on the *true* state and is scored on the clean S_d: gap-closed $= (S_{\text{HSM}}(\text{repair}(S, \hat{d}^*)) - S_{\text{HSM}}(S))/(1 - S_{\text{HSM}}(S))$. We report, for each noise level $\sigma \in \{0.00, 0.05, 0.10, 0.20\}$, both the agreement rate $\Pr[\hat{d}^* = d^*]$ and the mean gap-closed over $K = 100$ Monte Carlo trials per cross-phase pair. The clean ($\sigma = 0$) case recovers the exact decomposition above; the noisy cases characterise how much measurement budget the operator must invest to keep Alg. 2 strictly above the random-repair baseline.

3.7 Access-Attribute Dependence

Definition 2 (Access-Attribute-Dependent Method)**.** *A method M for index management is* access-attribute-dependent *if: (a) M constructs or maintains a physical data structure optimised for a fixed access-attribute set $A^* = (\mathcal{T}^*, \mathcal{C}^*)$; (b) when the observed access-attribute set $A(W_c) \neq A^*$, M must reconstruct its physical structure to remain optimal; and (c) that reconstruction incurs $\Theta(N \log N)$ cost.*

This definition applies universally—B-Trees, cracking, and all learned indexes satisfy it. HSM is not access-attribute-dependent: as a measurement function reading pre-aggregated window statistics, it remains logically unimpaired by any change in the physical access column, consistent with Codd's principle of physical data independence [46]. Theorem 3 formalises this property.

4 THEORETICAL ANALYSIS

We establish six core theorems supported by four lemmas. Lemmas 1–4 collect the foundational properties that justify the construction (metric axioms, parameter selection, complexity bounds, and a wavelet sublemma). Theorem 1 quantifies the wavelet approximation error of S_P. Theorem 2 proves the five HSM dimensions are minimal and sufficient. Theorem 3 unifies the economic break-even analysis with HSM's access-attribute independence under a single cost model. Theorem 4 establishes a stratified-Hoeffding concentration bound on the *intrinsic* detector quality (TPR, TNR) obtained by reframing HSM as a binary classifier of phase boundaries. Theorem 5 transports that band into a deployment-time certificate on the asymptotic speedup, parameterised by the operator's phase-mix prior π_w. Theorem 6 closes the framework with the convex-optimal window size N^*_{pts} that minimises total steady-state cost.

4.1 Supporting Lemmas

Lemma 1 (HSM Metric Properties)**.** *$\forall\, W_i, W_j, W_k \in \mathcal{W}$: $(P1)$ $S_{\text{HSM}}(W_i, W_j) \in [0, 1]$ (non-negativity and boundedness); $(P2)$ $S_{\text{HSM}}(W_i, W_i) = 1$ (identity of indiscernibles); $(P3)$ $S_{\text{HSM}}(W_i, W_j) = S_{\text{HSM}}(W_j, W_i)$ (symmetry); $(P4)$ $d_{\text{HSM}}(W_i, W_k) \leq d_{\text{HSM}}(W_i, W_j) + d_{\text{HSM}}(W_j, W_k)$ where $d_{\text{HSM}} = 1 - S_{\text{HSM}}$ (triangle inequality).*

Proof sketch. Each $d_i = 1 - S_i$ is an analytic metric; convex combination preserves boundedness (P1), identity (P2), symmetry (P3), and triangle inequality (P4). Full proof: Supplementary §I.

Lemma 2 (DWT Parameter Selection)**.** *Let $k = 4$ be the number of canonical workload states (volume regime $\times$ query-type dominance). The minimum SAX alphabet size that guarantees correct state discrimination under the maximum-entropy model with the two-sided burst extension satisfies $\alpha^* = 2k = 8$; under Quantile-SAX breakpoints, $P(\text{misclass.}) < 1/\alpha = 1/8$.*[4]

Lemma 3 (Tight Linear Complexity)**.** *$T_{\text{HSM}}(N_{\text{pts}}, N) = \Theta(N_{\text{pts}})$ in both upper and lower bound, independent of database cardinality N; space complexity is $\Theta(N_{\text{pts}})$. The lower bound*

4. The deployed pipeline uses $\alpha = k = 4$: DWT detail bands cD_1–cD_3 encode burst direction, leaving SAX on cA_3 to discriminate only the k canonical states; $\alpha^* = 2k$ is the SAX-only fallback (Supp. §II).

holds for any function $\mathrm{sim} : \mathcal{W} \times \mathcal{W} \to [0,1]$ *obeying P1–P4: such a function must inspect* $\Omega(N_{\mathrm{pts}})$ *elements of each input window, since otherwise an adversary can perturb an unread element to flip the value of* sim *while preserving P1–P4.*

Proof sketch. Upper: each dimension $O(N_{\mathrm{pts}})$ with no page access; total $\Theta(N_{\mathrm{pts}})$. Lower: adversary argument. Full proof: Supplementary §III.

Lemma 4 (Wavelet Coefficient Bound)**.** *For any signal* $q(t)$ *on* $[0, N_{\mathrm{pts}})$ *with* $\sup|q(t)| \le M$ *and its* L^2 *norm* $\|q\|_2 \le M\sqrt{N_{\mathrm{pts}}}$*, the level-L db4 DWT approximation* $\tilde{q}$ *satisfies* $\|q - \tilde{q}\|_2 \le C^L_{\mathrm{db4}}\, M\sqrt{N_{\mathrm{pts}}}$*, with* $C_{\mathrm{db4}} = \sqrt{2}/2 \approx 0.707$ *[25].*

4.2 Theorem 1: Wavelet Pattern Preservation

Theorem 1 (Wavelet Pattern Preservation)**.** *Let* $q(t)$ *satisfy* $\sup|q(t)| \le M$*. The level-L db4 DWT used by* S_P *introduces pattern-similarity error bounded by*

$$\varepsilon(M, N_{\mathrm{pts}}, L) \;=\; \frac{2\, C^L_{\mathrm{db4}}\, M}{\sqrt{N_{\mathrm{pts}}}}, \qquad C_{\mathrm{db4}} = \frac{\sqrt{2}}{2} \approx 0.707, \quad (7)$$

which is strictly decreasing in both L *and* N_{pts}*. Equation* (7) *is a closed-form expression in operator-measurable quantities.*

Proof sketch. Follows from Lemma 4 and the normalised FastDTW score sensitivity. Full proof: Supplementary §V. At the v2 default ($L = 3$, $N_{\mathrm{pts}} = 100$), $\varepsilon \le 0.071\, M$.

4.3 Theorem 2: Minimal Sufficient Dimensions

$\forall\, d \in \{S_R, S_V, S_T, S_A, S_P\}$, let HSM_{-d} denote HSM with dimension d removed and its weight redistributed equally over the remaining four dimensions.

Theorem 2 (Minimal Sufficient Dimensions)**.** (Necessity.) $\forall\, d$, $\exists\, (W_A, W_B), (W_C, W_D) \in \mathcal{W}$ *such that* (i) $S_d(W_A, W_B) = S_d(W_C, W_D)$ *(d is uninformative on this pair), and* (ii) $|S_{\mathrm{HSM}}(W_A, W_B) - S_{\mathrm{HSM}}(W_C, W_D)| \ge w_d$*. Hence* HSM_{-d} *is a strictly weaker discriminator than* HSM*.* (Sufficiency.) *The five dimensions* $(S_R, S_V, S_T, S_A, S_P)$ *collectively span rank, magnitude, angular, attribute-set, and temporal information; no two are linearly equivalent on* $\mathcal{W}$*.* (Minimality.) *No four-dimensional subset of these five is sufficient: removing any one violates at least one witness pair above. Full proof with explicit witnesses: Supplementary Section VI.*

Empirical caveat (global sufficiency vs. per-workload utility). Theorem 2 establishes *global* sufficiency and minimality: the witness construction guarantees that no four-dimensional subset suffices on the joint workload space. This does *not* imply every dimension is equally informative on every single workload. On TPC-H, $\Delta S_P \approx 0$ across all SFs (vs. $\Delta S_T \approx 0.53$, $\Delta S_A \approx 0.56$), so (S_T, S_A) dominate phase separation; we retain S_P because (i) it carries unique intra-window shape information required by the burst witness pair (necessary for global sufficiency in Theorem 2), and (ii) workloads with temporal periodicity make S_P informative (14 % contribution on burst A8d). Per-dimension utility across five workloads is quantified in the Supplementary (Dimensional-Utility Table).

4.4 Theorem 3: Economic Optimality of HSM Gating

The cost model uses system-measurable constants a, b, f, g, c (full parameter table in the Supplementary, Cost-Model Parameters).

Theorem 3 (Economic Optimality of HSM Gating)**.** *Under this cost model, define the break-even query volume* $Q_{\min}(N) = (aN\log N + b)/(fN - g\log N)$*. Then:* (i) $Q_{\min}(N) = \Theta(\log N)$ *as* $N \to \infty$*, and* $C_{\mathrm{comp}}/C_{\mathrm{create}} \to 0$*;* (ii) *for any next-window query volume* $Q > Q_{\min}(N)$*, the optimal HSM decision threshold is* $\theta^*(N, Q) = 1 - Q_{\min}(N)/Q$*, derived by equating the expected gated and naive costs;* (iii) *HSM's per-window computation cost is* $T_{\mathrm{HSM}} = \Theta(N_{\mathrm{pts}})$ regardless *of the access attribute* $A(W)$*, by Lemma 3;* (iv) *for any access-attribute-dependent method* M*, an access-attribute transition forces* $\Theta(N\log N)$ *reconstruction cost on* M *but only* $O(|A|) = O(1)$ *marker update cost on HSM, so over* τ *transitions HSM saves* $\Omega(\tau \cdot N\log N)$*;* (v) *consequently* $T_{\mathrm{HSM}}/T_M \to 0$ *as* $N \to \infty$*.*

Proof sketch. (i)–(ii): limit analysis and cost minimisation in θ. (iii)–(v): substitute Lemma 3; ratio vanishes because N_{pts} is constant in N. Full proof: Supplementary §VII.

A structural comparison against B-tree, cracking, learned indices, and recent workload-similarity methods (BALANCE, Indexer++, ShapeShifter, Brucato) appears in the Supplementary (Cost-Model Parameters; Structural Comparison): HSM is the only method that simultaneously satisfies P1–P4, exposes five orthogonal dimensions, supplies a continuous score with a closed-form threshold, requires no labelled training data, and amortises advisor cost to $O(1)$ per window.

4.5 Theorem 4: Detector-Quality Concentration via Stratified Hoeffding

We reformulate HSM as a binary classifier of phase boundaries. For a threshold $\theta \in [0,1]$, HSM decides RETAIN on a window pair iff $S_{\mathrm{HSM}} \ge \theta$. Two intrinsic quantities characterise the detector independently of how often phase changes actually occur in the deployment workload: the true-positive rate $\mathrm{TPR}(\theta) = \Pr[S_{\mathrm{HSM}} \ge \theta \mid \text{within-phase}]$ and the true-negative rate $\mathrm{TNR}(\theta) = \Pr[S_{\mathrm{HSM}} < \theta \mid \text{cross-phase}]$. Theorem 4 establishes a non-asymptotic certificate on these two rates via *stratified* sampling, which yields tighter bands than the single-rate Hoeffding bound used in earlier HSM work.

Theorem 4 (Detector-Quality Concentration via Stratified Hoeffding)**.** *Let* $\hat{\mathrm{TPR}}_{m_w}$ *and* $\hat{\mathrm{TNR}}_{m_c}$ *be the empirical rates measured from* m_w *i.i.d. within-phase pairs and* m_c *i.i.d. cross-phase pairs respectively. Each indicator is bounded in* $[0,1]$*. Then for any* $\delta \in (0,1)$*, with* $\eta(m,\delta) = \sqrt{\log(4/\delta)/(2m)}$*,*

$$\Pr\Big[\,|\hat{\mathrm{TPR}}_{m_w} - \mathrm{TPR}| \le \eta(m_w,\delta) \;\wedge\; |\hat{\mathrm{TNR}}_{m_c} - \mathrm{TNR}| \le \eta(m_c,\delta)\,\Big] \;\ge\; 1-\delta \quad (8)$$

i.e. the simultaneous confidence band $[\hat{\mathrm{TPR}}_{m_w} - \eta_w,\ \hat{\mathrm{TPR}}_{m_w} + \eta_w] \times [\hat{\mathrm{TNR}}_{m_c} - \eta_c,\ \hat{\mathrm{TNR}}_{m_c} + \eta_c]$ *covers the true detector-quality pair* $(\mathrm{TPR}, \mathrm{TNR})$ *with probability at least* $1 - \delta$*. The decisive advantage of stratification over a single Hoeffding bound on the mixture* $p_{\mathrm{retain}} = \pi_w \mathrm{TPR} + (1-\pi_w)(1-\mathrm{TNR})$ *is* operational identifiability*: the two intrinsic quantities* $(\mathrm{TPR}, \mathrm{TNR})$ *are recovered separately, so Theorem 5 can transport them into a deployment certificate at* any *operator-supplied* π_w*, whereas the mixture*

rate confounds detector quality with workload prior. Variance-based concentration bounds (Bernstein/Bennett) additionally give a strictly tighter band under stratification.

Proof sketch. Apply Hoeffding independently per stratum at $1-\delta/2$ and union-bound. Full proof: Supplementary §VIII. At SF = 3.0 (worst-case among evaluated scales; $\theta = 0.75$, $m_w = m_c = 160$, $\delta = 0.05$): $\eta \approx 0.117$, with $\hat{\mathrm{TPR}} = 0.986$ and $\hat{\mathrm{TNR}} = 0.970$, giving $\mathrm{TPR} \in [0.869, 1.000]$, $\mathrm{TNR} \in [0.853, 1.000]$ at 95% confidence—inputs to Theorem 5.

Sensitivity as a design feature, and θ as an operating point. The imperfect empirical rates ($\hat{\mathrm{TPR}} = 0.986$, $\hat{\mathrm{TNR}} = 0.970$) are a direct consequence of the metric construction (Lemma 1): the angular transforms d_R, d_T satisfy the triangle inequality *precisely because* they are locally linear near identity, so small distributional deviations map to proportional metric distances rather than being compressed into a plateau. HSM therefore reacts to genuine within-phase sampling variance, the desirable behaviour of a metric-compliant detector. Consequently θ is an *operating point* on the detector's ROC curve rather than a universal constant: Supplementary §XIV reports that the Youden-optimal θ^* varies across engines and workloads (TPC-H, JOB/IMDB, SDSS, OLTP), so operators can raise θ on highly regular workloads or lower it for early warning on subtle drift. The default $\theta = 0.75$ is the Youden-optimum on aggregated TPC-H+JOB calibration and dominates periodic rebuild at every evaluated SF.

4.6 Theorem 5: Operational Speedup with Phase-Mix Prior

The original asymptotic gating result of HSM treated p_{stable} as a single opaque scalar. Once HSM is reformulated as a binary classifier, p_{stable} decomposes into two independent factors: an *intrinsic* detector quality $(\mathrm{TPR}, \mathrm{TNR})$ certified by Theorem 4, and a *deployment-dependent* phase-mix prior $\pi_w = \Pr[\text{a window is within-phase}]$ that the operator estimates from their workload's change-point rate. Theorem 5 transports the Theorem 4 confidence band into a parameterised speedup curve.

Theorem 5 (Operational Speedup with Phase-Mix Prior)**.** *Let $\mathcal{A}$ be any index-management system with $T_{\mathcal{A}}(N) = \Omega(N \log N)$, and let $p_{\mathrm{stable}}(\pi_w, \mathrm{TPR}, \mathrm{TNR}) = \pi_w \mathrm{TPR} + (1-\pi_w)(1-\mathrm{TNR})$. Define $\mathrm{Speedup}(N) = \mathrm{Cost}_{\mathrm{naive}}(N)/\mathrm{Cost}_{\mathrm{gated}}(N)$. Then:*

- *(i)* (Asymptotic limit.) $\mathrm{Speedup}(N) \to 1/(1-p_{\mathrm{stable}})$ *as $N \to \infty$; the HSM overhead ratio $\to 0$ and absolute savings grow without bound.*
- *(ii)* (Convergence rate.) $\left|\mathrm{Speedup}(N) - \frac{1}{1-p_{\mathrm{stable}}}\right| = O(T_{\mathrm{HSM}}/T_{\mathcal{A}}(N)) = O(N_{\mathrm{pts}}/(N \log N))$ *since $T_{\mathrm{HSM}} = \Theta(N_{\mathrm{pts}})$ (Lemma 3).*
- *(iii)* (Monotone transport.) *The map $(\mathrm{TPR}, \mathrm{TNR}) \mapsto p_{\mathrm{stable}}(\pi_w, \cdot, \cdot)$ is non-decreasing in both arguments at every fixed $\pi_w \in [0,1]$, and $p \mapsto 1/(1-p)$ is strictly increasing on $[0,1)$.*
- *(iv)* (Deployment certificate.) *Conditional on the Theorem 4 band $(\mathrm{TPR}, \mathrm{TNR}) \in \mathcal{B}_\delta$ and any operator estimate π_w, with probability at least $1-\delta$,*

$$\mathrm{Speedup}_\infty \;\geq\; \frac{1}{1-\pi_w(\hat{\mathrm{TPR}}-\eta_w)-(1-\pi_w)(1-(\hat{\mathrm{TNR}}+\eta_c))},$$

$$\mathrm{Speedup}_\infty \;\leq\; \frac{1}{1-\pi_w(\hat{\mathrm{TPR}}+\eta_w)-(1-\pi_w)(1-(\hat{\mathrm{TNR}}-\eta_c))}.$$

Proof sketch. (i)–(ii): geometric series yields the Amdahl-style limit. (iii): direct partial derivatives. (iv): monotone composition transports the Theorem 4 band to a closed-form speedup interval. Full proof: Supplementary §IX. Numerical values at SF = 3.0 (worst case; $\hat{\mathrm{TPR}} = 0.986$, $\hat{\mathrm{TNR}} = 0.970$) are reported once in §V.A14 (Theorem-5 certificate and the π_w-parameterised curve); Supplementary §XII gives the full per-SF tables and the scale analysis figure.

Applicability to state-of-the-art advisors. The premise $T_{\mathcal{A}}(N) = \Omega(N \log N)$ is satisfied by *every* published advisor we surveyed (Table 2): each touches $\Omega(N)$ rows to evaluate, build, or amortise a candidate index. Theorem 5 therefore applies *orthogonally*—HSM inserts a decision (*when*) layer in front of any index-selection (*what*) advisor without modifying its internals. End-to-end validation uses Dexter [10] and `index_advisor` [12]; the same protocol extends to GPU-trained advisors (BALANCE, MFIX, Indexer++).

TABLE 2
Applicability of Theorem 5 to recent index advisors. HSM gates the host advisor without modifying its internals; the $\Omega(N \log N)$ premise is satisfied by every published system because each touches $\Omega(N)$ rows during cost-model evaluation or physical index construction.

Advisor	Layer	Why $T_{\mathcal{A}} = \Omega(N \log N)$
Academic (research prototypes; GPU/training required)		
BALANCE [5]	Index sel.	Cost-model scan over N rows per candidate; sorted-merge eval.
MFIX [6]	Index sel.	Multi-fidelity Bayesian: each fidelity builds/probes physical indexes ($\Theta(N \log N)$).
Indexer++ [7]	Index sel.	DRL episode includes B-tree construction passes ($\Theta(N \log N)$ per build).
ShapeShifter [8]	Index sel.	Workload-similarity-driven re-tuning; physical rebuilds dominate.
DBMind [13]	Index sel.	RL-based; openGauss (PostgreSQL fork); $\Theta(N \log N)$ per candidate physical build.
Deployed (open source, vanilla PostgreSQL)		
Dexter [10]	Index sel.	What-if via HypoPG + cost-model over relation stats ($\Omega(N)$ amortised); widely deployed.
`index_advisor` [12]	Index sel.	HypoPG-based SQL-driven what-if; cost model iterates over relation statistics.
DTA [4]	Index sel.	Classical what-if API; iterates over per-candidate cost estimates ($\Omega(N)$).
HSM (this paper)	**Decision (when)**	**$\Theta(N_{\mathrm{pts}})$, $N_{\mathrm{pts}} \ll N$; gates any of the above.**

4.7 Theorem 6: Optimal Window Size

Theorem 5 parameterises deployment speedup by phase-mix prior π_w and detector quality $(\mathrm{TPR}, \mathrm{TNR})$, but treats window size N_{pts} as an exogenous constant. In practice N_{pts} is itself a design choice governed by two opposing pressures: *approximation*—too few points erases the wavelet-visible phase signal, reducing TPR—and *measurement cost*—too many points inflates $\Theta(N_{\mathrm{pts}})$ HSM overhead and per-rebuild advisor cost. Theorem 6 closes the framework by giving the unique N^*_{pts} that minimises total steady-state cost.

Theorem 6 (Optimal Window Size)**.** *Consider the per-window expected cost*

$$\mathcal{C}(N_{\text{pts}}) = \underbrace{aN_{\text{pts}}\log N_{\text{pts}}}_{\text{DWT/SAX/DTW}} + \underbrace{\lambda(f_{\text{N}} - g\log N_{\text{pts}})}_{\text{miss-detect rebuild}} + \underbrace{\frac{M^2 C_{\text{db4}}^{2L}}{N_{\text{pts}}^2}}_{\text{wavelet approx.}} \quad (9)$$

where (i) $aN_{\text{pts}}\log N_{\text{pts}}$ is a conservative upper bound on the measurement cost that accommodates kernel variants beyond the default FastDTW-based pipeline of Lemma 3 (which establishes the tight $\Theta(N_{\text{pts}})$ bound, a strict special case of $aN_{\text{pts}}\log N_{\text{pts}}$); using this conservative form makes Theorem 6 robust to slower DWT/DTW variants, and $a > 0$ is the per-point HSM constant; (ii) the miss-detection term is a first-order expansion of $(1-\text{TPR}(N_{\text{pts}}))$ around the operating point using the wavelet pattern-preservation bound of Theorem 1 with $f_{\text{N}}, g > 0$ empirical constants, and λ the expected per-miss advisor-rebuild cost; (iii) $M^2 C_{\text{db4}}^{2L}/N_{\text{pts}}^2$ is the squared ℓ_2 reconstruction bound of Theorem 1 with $M = \max_n |x[n]|$, L the DWT depth, and $C_{\text{db4}} = \sqrt{2}/2 \approx 0.707$ the Daubechies-4 attenuation constant of Lemma 4. Then $\mathcal{C}$ admits a unique global minimum N_{pts}^* characterised by the stationarity condition $\mathrm{d}\mathcal{C}/\mathrm{d}N_{\text{pts}} = 0$, which, after rearrangement, yields the implicit fixed-point equation

$$N_{\text{pts}}^* = \sqrt{\frac{8\, C_{\text{db4}}^{2L}\, M^2\, a\, N_{\text{pts}}^* \log N_{\text{pts}}^*}{\lambda(f_{\text{N}} - g\log N_{\text{pts}}^*)}} \quad \textit{(solve numerically)} \quad (10)$$

in a single scalar N_{pts}^*. $\mathcal{C}$ is strictly convex in N_{pts} for all operating regimes with $g\log N_{\text{pts}} < f_{\text{N}}$ (i.e. the linearised TPR approximation is valid), so N_{pts}^* is the unique root of $\mathrm{d}\mathcal{C}/\mathrm{d}N_{\text{pts}} = 0$ in $[N_{\text{min}}^{\text{pts}}, N_{\text{max}}^{\text{pts}}]$, where $N_{\text{min}}^{\text{pts}} \geq 8$ ensures sufficient sample size for rank-based statistics (S_R, Spearman ρ) and SAX discretisation.[5] *Bisection or fixed-point iteration on Eq.* (10) *converges in $O(\log(1/\epsilon))$ steps for any tolerance $\epsilon > 0$.*

Proof sketch. Differentiating (9), $\mathrm{d}\mathcal{C}/\mathrm{d}N_{\text{pts}} = a(\log N_{\text{pts}} + 1) - \lambda g/N_{\text{pts}} - 2M^2 C_{\text{db4}}^{2L}/N_{\text{pts}}^3$; the second derivative $\mathrm{d}^2\mathcal{C}/\mathrm{d}N_{\text{pts}}^2 = a/N_{\text{pts}} + \lambda g/N_{\text{pts}}^2 + 6M^2 C_{\text{db4}}^{2L}/N_{\text{pts}}^4 > 0$ for all $N_{\text{pts}} > 0$ and positive parameters, establishing strict convexity and hence a unique global minimum. Solving the stationarity condition gives (10). Full proof: Supplementary §X.

Corollary 1 (Cross-Scale Consistency of N_{pts}^*)**.** *Assume the rebuild cost scales as $\lambda(N) = \Theta(N\log N)$ (the premise $T_{\mathcal{A}}(N) = \Omega(N\log N)$ of Theorem 5), and assume M, a, f_{N}, g, L are fixed design constants. Then the convex-optimal N_{pts}^* of Theorem 6 is invariant in N to leading order: $N_{\text{pts}}^*(N_1)/N_{\text{pts}}^*(N_2) \to 1$ as $\min(N_1, N_2) \to \infty$. Hence a single operator-chosen N_{pts}^* remains within one dyadic step of optimal across scales.*

Proof. Both numerator and denominator of (10) scale as $\Theta(N\log N)$ under the assumption; ratios cancel and the square-root preserves the scaling. Full derivation: Supplementary §X.

Empirical validation. On the window-size sweep A17, the empirical minimum of $\mathcal{C}(N_{\text{pts}})$ occurs at $N_{\text{pts}}^* = 50$ on TPC-H SF = 0.2 *and* SF = 3.0 (DR = 1.85, Youden $\hat{J} = 1.00$ with 95% Hoeffding lower bound $\hat{J}_{\text{LB}} \geq 0.77$ at $m_w = m_c = 160$, $\delta = 0.05$), and at $N_{\text{pts}}^* = 100$ on JOB/IMDB SF = 1.0 ($\hat{J} = 1.00$, same LB). All three minima match the Theorem 6 prediction within one dyadic step, and the 15× change in N (SF 0.2 vs. SF 3.0) leaves N_{pts}^* unchanged at 50, consistent with Corollary 1. To rule out cache-state bias, every A17 timing used `DISCARD ALL` plus a three-query warm-up, and the $\Theta(N\log N)$ fit of A12 ($\beta = 1.177$, $R^2 = 0.998$) is reproduced identically on cold and warm runs. The practical consequence is that the default $N_{\text{pts}} = 32$ used throughout Sections V–VI is within 1.5× of optimal cost, and operators may increase to $N_{\text{pts}}^* \in [50, 100]$ for maximum per-window discrimination when rebuild cost λ is large—without re-sweeping N_{pts} for each new scale factor.

5. The real-valued solution is rounded to the nearest admissible integer $\lceil N_{\text{pts}}^* \rceil$ or $\lfloor N_{\text{pts}}^* \rfloor$, whichever yields lower $\mathcal{C}$; convexity guarantees the rounded value is within one unit of optimal.

5 Experimental Evaluation

5.1 Experimental Setup

Hardware and software environment. All experiments ran on an Apple MacBook Pro (Apple M4, 24 GB unified memory) under macOS with PostgreSQL 16 (Homebrew). PostgreSQL was production-tuned: `shared_buffers`=6 GB, `work_mem`=256 MB, `effective_cache_size`=18 GB, `maintenance_work_mem`=2 GB, `wal_buffers`=16 MB [47], [48]. Default settings would render large-SF queries I/O-bound; production tuning ensures $T_A(N)$ reflects realistic operational conditions and directly validates Theorem 4 scaling. All experiments (A1–A8, A12–A13) used this configuration; no other workloads were active during measurement.

Workload trace. A 32-window TPC-H SF = 0.2 trace (8 tables, ≈1.2M rows) uses dbgen v3.0.0 with four phases of 8 windows each: Reporting (scan-heavy aggregations), Join-Heavy (multi-table joins), Procurement (supply-chain), Customer Analysis; ground-truth boundaries at W7→W8, W15→W16, W23→W24. Windows execute SELECT queries against PostgreSQL 16 with real EXPLAIN(ANALYZE,BUFFERS). Noise model: (N1) window size ∼Poisson($\lambda = 20$) $\in [8, 40]$; (N2) per-window query-type probabilities re-sampled from Dirichlet($\alpha = 8w$); (N3) execution-time LogNormal($\sigma = 0.35$) jitter; (N4) DML keys ∼Zipf($s = 1.2$); (N5) 5% phase leakage. Ten seeds (block_seed 7120–8020, step 100) on freshly loaded databases; ICC(2,1) [49] computed.

Experimental tiers. Tier 1 (A1–A8): SF = 0.2 for reproducible, scale-independent theorem validation. Tier 2 (A12, Section V): $T_A(N)$ across SF $\in \{0.2, 1.0, 3.0\}$ to validate $\Theta(N\log N)$ rebuild scaling and Theorem 4 gating efficiency. Tier 3 (A-CE, Section 5.6): cross-engine validation on MongoDB 7, demonstrating DBMS-agnostic HSM with the document-store extractor (Section 3.3).

The PostgreSQL workload comprises four query-mix phases (Reporting, Join-Heavy, Customer, Procurement; window groups 1–8, 9–16, 17–24, 25–32) drawing from TPC-H queries Q1–Q22 with Poisson-sampled window sizes (mean 19.5, range 16–20). The MongoDB workload comprises four document-store phases (Edge, Geo, Text, Review) on a single polymorphic collection `combined_clean` (5 M documents, five labelled types) with ground-truth drift at W7, W13, W19 and fixed window size $N_{\text{pts}} = 20$. Full per-phase mappings appear in the Supplementary (Workload Phase Mappings).

5.2 Research Questions

Experiments are organised in three parts: Part I primary validation (A1–A8), Part II extended validation (A9–A14), and Part III cross-engine validation on MongoDB (A-CE1–3). The full RQ→theorem map appears in the Supplementary (Research-Question Map). HSM overhead $S_o = 1.09$ ms (0.25 % of mean query time) is uniform across A1–A7 because N_{pts} is held constant.

5.3 A1–A7: Dimension Validation, Composite Score, and Complexity

A1–A4. All four metric axioms (Lemma 1) hold on 280 within-phase pairs with zero violations (within-phase means: $S_R = 0.942$, $S_V = 0.995$, $S_T = 0.988$). At phase boundaries S_A drops 63.2 % (mean 0.364 vs. 0.990 within). Single-dimension ablation: removing S_T reduces SDSS DR by 14.4 % and TPC-H DR by 11.1 %; removing S_A reduces TPC-H DR by 7.2 %; removing S_R reduces TPC-H DR by 6.5 % (full table in Supplementary, Dimension-Ablation Study). **A5.** Over ten seeds the all-pairs protocol yields 1,120 within-phase and 3,840 cross-phase comparisons; DR = 1.624 (95 % CI [1.613, 1.636]), $p < 0.001$, $|r| = 0.999$, ICC(2,1) = 0.952. The composite S_{HSM} trajectory drops below the $\theta = 0.75$ gate only at the genuine boundary W7→W8 (full per-window trajectory in the Supplementary, A5–A7 Composite Score and Pipeline Complexity). **A6.** Algorithm 1 fires REBUILD only at W7→W8 ($S_{\text{HSM}} = 0.74$); W15→W16 and W23→W24 remain above threshold across all ten seeds. The resulting phase-stability estimate $p_{\text{stable}} = 0.875$ implies, by Theorem 5, an asymptotic $8\times$ speedup over always-on advisor invocation at this operating point. **A7.** Log-log OLS yields slope = 0.935 ($R^2 = 0.997$), confirming $\Theta(N_{\text{pts}})$ scaling (Lemma 3; Supplementary, A12/A14 Scale Sensitivity).

5.4 A8: Generalizability Assessment

HSM generalises across four independent workloads with no re-tuning of weights or threshold. SDSS SkyServer (16,000 queries, ten seeds): DR = 1.310 ([1.304, 1.315]), S_T dominant (-14.4 % ablation). JOB/IMDB [50]: DR = 1.250 ([1.246, 1.255]), $|r| = 0.997$, S_V+S_T dominant on complexity-tier phases. pgbench TPC-B: DR = 2.008 ([1.993, 2.023]), S_R+S_V dominant (write-heavy). Burst workload: DR = 1.093 ([1.086, 1.101]), S_P dominant (Limitation iv). Every dimension contributes $\Delta > 0.10$ in at least one workload, validating Theorem 2 (per-dimension utility table in the Supplementary, A8: Generalizability (Full Results)).

5.5 A9–A13: Overhead, Robustness, and Decision-Layer Cost

A9 (Overhead). HSM consumes 1.09 ms of CPU per transition (0.25 % of mean query time, zero query-latency impact, 50 KB per-window buffer, 919 trans/s throughput). S_P (wavelet+SAX+FastDTW) dominates at 526 μs; the remaining four dimensions sum to $< 600\,\mu$s, leaving total compute $396\times$ below mean query execution time, so HSM does not become the critical path on the workloads tested. **A10 (Robustness).** Independent ± 60 % perturbations of the five noise parameters N1–N5 leave DR within ≤ 6.1 % of nominal for N1–N4. Phase leakage (N5) is the only parameter with measurable effect (± 13 %), as expected since N5 directly contaminates the labels HSM discriminates; DR remains above unity throughout. **A10b (Weight Sensitivity).** Across all ± 0.05 single-weight perturbations of $\mathbf{w}_0 = (0.25, 0.20, 0.20, 0.20, 0.15)$, the maximum DR change is 5.5 % (TPC-H 1.2–3.4 %; SDSS 0.4–7.1 %), confirming a broad flat optimum. **A11 (Break-even, Theorem 3).** The break-even query volume of Theorem 3 crosses HSM cost at $N_{\text{pts}} \approx 17{,}000$, three orders of magnitude above the measured OLAP per-window mean of 18 queries; this corresponds to a $\approx 26\times$ payoff per averted advisor call. Full noise/weight robustness grids and the break-even cost-model parameters appear in the Supplementary (Cost-Model Parameters; Cross-Workload θ^* Generalisability).

A13 (Decision-Layer Cost on TPC-H, Theorem 4). On a 10-block RCB across SF $\in \{0.2, 1.0, 3.0\}$ with PostgreSQL 16, HSM-gated retains 61.9 ± 1.2 % of baseline throughput while always-on retains only 21.3 ± 1.7 % and periodic ($K = 3$) retains 43.4 ± 2.0 %. Equivalently, HSM fires only 4.6 ± 1.4 times per 24-window block (vs. 24 and 8 for always-on and periodic); Table 4 rows A13/A13c report the corresponding invocation-savings percentages. The decision-layer ratios are $2.92 \pm 0.18\times$ over always-on and $1.43 \pm 0.04\times$ over periodic, and they are scale-invariant across the $15\times$ data-volume range (CV 6.2 % and 2.5 %). The measured $\hat{p}_{\text{stable}} = 0.808$ yields a Theorem 5 bound of $5.22\times$; the realised $2.92\times$ falls short by exactly the $O(N_{\text{pts}}/(N \log N))$ rebuild-amortisation gap of part (ii). The relevant comparison for a decision layer is against *other* invocation policies, not against no-advisor (which incurs zero rebuild cost on a workload whose schema already fits the trace). Full per-SF tables and the throughput figure appear in the Supplementary (A13 Decision-Layer Cost).

A13b–c (Gate sensitivity and cross-workload speedup bound). The TPC-H gate margin is $z = 3.40$–4.15σ ($d' > 12$, detection probability 100 %); $z > 3$ is the deployment-ready rule (Supplementary, Gate Sensitivity). Replaying Theorem 5 on four additional workloads (OLTP, burst, JOB/IMDB, SDSS) at each workload's empirical $\hat{p}_{\text{stable}}$ (CSV of record: `results/t5_validation.csv`): burst at $\theta = 0.75$ attains $80.8\,\%/44.4\,\%$ savings under a $5.20\times$ bound; SDSS at $\theta^* = 0.6325$ (Youden) attains $75.1\,\%/25.2\,\%$ under a $4.01\times$ bound; JOB/IMDB at $\theta^* = 0.7275$ ($n = 20$, sample-limited) attains $95.0\,\%/85.7\,\%$ under a $20.0\times$ bound. OLTP correctly self-deactivates: pgbench drifts on essentially every window, so $\hat{p}_{\text{stable}} \to 0$ and the bound itself collapses to $1\times$—no gating policy can save work on a workload that never stabilises. Per-workload TPR/FPR appears in the Supplementary (Cross-Workload θ^* Generalisability).

A13d (End-to-End Validation with Deployed Advisors). We re-run the A13 protocol on a synthetic burst workload while replacing the advisor stub with two deployed open-source PostgreSQL advisors: Dexter [10] and Supabase `index_advisor` [12] (both HypoPG-based, §II.A). PostgreSQL 16 with HypoPG 1.4; pgbench scale 10 ($N = 10^6$, giving $Q_{\min}(10^6) \approx 122$). Each policy $\times$ advisor combination executes ten paired blocks against a no-advisor baseline; the primary metric is the Cost-Effective Timing Savings $\text{CETS}_{\text{base}} = 1 - C_{\text{HSM}}/C_{\text{base}}$ of Theorem 5 part (i), measured rather than predicted. We evaluate three operating points on the Theorem 3 economic frontier: small-Q ($Q = 35 < Q_{\min}$,

theory predicts $\theta^* = 0$); mid-Q ($Q = 700$, $\theta^* = 0.826$, exactly one of two ground-truth boundaries is economic); and large-Q ($Q = 3000$, $\theta^* = 0.959$ in theory, clamped to the within-phase noise floor $S_{\text{floor}} \approx 0.875$). Table 3 reports wall-clock, advisor cost, trigger count, and precision; brackets give 95 % bootstrap CIs ($B = 10\,000$, paired within-block).

Receipt for Theorem 3: small-Q retention. At $Q = 35$ ($< Q_{\min}$), HSM-gated under both advisors fires *zero* times over the entire 10-block run, matching $\theta^*(N, Q) = 0$ exactly. Wall-clock is within 8% of the no-advisor baseline (1.075 vs. 0.996 s, non-overlapping CIs but a negligible absolute gap of ≈ 80 ms) with p95 latency within ≈ 1 ms—the HSM decision layer is effectively free. Against the scheduled policies, $\text{CETS}_{\text{always}}$ is 66.4–68.0% and $\text{CETS}_{\text{periodic},K=2}$ is 44.0–51.8%: because no advisor call is economic, every trigger the scheduled baselines issue is wasted work. Trigger precision for the scheduled policies is correspondingly low ($P_{\text{strict}} = 0.100$ at periodic $K = 2$, 0.095 at always-on), since both spend advisor cycles on windows that Theorem 3 proves are sub-economic.

Receipt for Theorem 3: mid-Q selectivity. At $Q = 700$ with $\theta^* = 0.826$, HSM-gated fires exactly once per run in every one of the 10 blocks, yielding $P_{\text{strict}} = 1.000$ with a degenerate $[1.000, 1.000]$ bootstrap interval, indicating that the gate both selects the economically profitable transition (BURST_ALT $\rightarrow$ BURST_GRP, $S = 0.781$, $Q_i = 557 < 700$) and correctly *skips* the shallower one ($S = 0.892$, $Q_1 = 1130 > 700$). Versus always-on, HSM-gated saves 66.5% (Dexter) and 71.1% (`index_adv.`) of wall-clock; versus periodic $K = 2$, 50.7% and 53.2%. Precision is 100×–200× the scheduled baselines (0.005 always-on, 0.010 periodic $K = 2$). Critically, p95 query latency under HSM-gated is *equal to or better than* the no-advisor baseline (Dexter: 42.59 vs. 42.69 ms; `index_adv.`: 28.80 vs. 42.69 ms with non-overlapping CIs—Supabase's `index_advisor` actually reduces tail latency below baseline via its one economic rebuild). This confirms HSM's skip decisions impose zero user-observable penalty (§5.5 *latency-on-skip* metric).

Practical θ bound: within-phase noise floor. Applying Theorem 3's $\theta^*(N, Q) = 1 - Q_{\min}/Q$ formula naively at $Q = 1500$ yields $\theta^* = 0.919$, which over-fires every window because the within-phase signature in our synthetic burst exhibits a *noise floor* at $S_{\text{floor}} \approx 0.875$ (random `aid`-sample variability). In general, the usable threshold is bounded below by the stable-phase distribution: $\theta^*_{\text{eff}}(N, Q) = \min\big(\theta^*(N, Q),\, S_{\text{floor}}(W)\big)$, where $S_{\text{floor}}(W)$ is the 5th-percentile within-phase similarity at window size W. The $Q = 700$ operating point respects this bound ($0.826 < 0.875$); the $Q = 1500$ operating point does not and is therefore excluded from the main table. For workloads whose $Q_{\min}/(1 - S_{\text{floor}}) > Q_i$ for every true transition i, HSM-gated's economic recall is necessarily $\leq |\{i : Q > Q_i\}| \,/\, |\mathcal{T}|$—i.e. HSM catches every transition that is economic *and* separable from within-phase drift. Here that yields economic-recall $1/1$ at $Q = 700$ and strict-recall $1/2$ (the second transition being correctly ignored per Theorem 3). We return to this in §6(iv).

B1 receipt: clamping prevents over-firing at large Q. At $Q = 3000$, Theorem 3 predicts $\theta^* = 0.959$, an aggressive bound. The within-phase HSM signature exhibits a noise floor $S_{\text{floor}} \approx 0.875$ on this workload, so the clamp $\theta_{\text{eff}} = \min(0.959,\, 0.875) = 0.875$ yields exactly 1 trigger per block at precision 1.000. Without the clamp, the naive $\theta = 0.959$ trigger fires 601 times per block (a 601× increase in advisor invocations) at precision 0.003. Wasted advisor cycles scale with advisor cost: 9.3× more advisor time on Supabase `index_advisor` (4.66 s vs. 0.50 s) and 325× more on Dexter (58.5 s vs. 0.18 s). Wall-clock impact depends on the advisor: Dexter degrades from 87 s to 146 s, while `index_advisor` actually completes faster naive (37 s vs. 61 s) because its lightweight recommend-only operations let early-built indexes amortise. The headline value of clamping is therefore precision: it filters the 600 false-positive triggers without trading on wall-clock for fast advisors. The Supplementary (Cross-Workload θ^* Generalisability) separately confirms that the angular S_R and S_T dimensions are amplitude-invariant ($k \in \{0.5, 1, 2, 5, 10\}$), with HSM floor in $[0.695, 0.875]$ across workloads (B4 receipt).

TABLE 3
End-to-end wall-clock validation on burst workload with deployed PostgreSQL advisors (Dexter, Supabase `index_advisor`), two operating points. Block medians with 95 % bootstrap CIs in brackets ($B = 10\,000$, paired within-block resamples, 10 blocks each variant); periodic $K = 2$ is the realistic scheduled baseline ($K \in \{3, 4\}$ in Supplementary §A13d Burst End-to-End: Extended Periodic Comparators). t_{wall} is total run wall-clock including advisor cost; t_{adv} is advisor-only; CETS is Cost-Effective Timing Savings (higher is better); P_{strict} is trigger precision (fraction of gate fires that land on a ground-truth phase boundary); p_{95}^{lat} is per-query p95 latency (ms). Raw CSVs in `results/burst_end_to_end/burst_small_q_results.csv` and `results/burst_end_to_end/burst_mid_q_results.csv`.

Variant	Policy (Advisor)	t_{wall} (s)	t_{adv} (s)	Calls	$\mathrm{CETS}_{\mathrm{always}}$	P_{strict}	p_{95}^{lat} (ms)
Small-Q: $Q = 35$ ($Q < Q_{\min}$), $\theta = 0.750$, 10 blocks							
	baseline	0.996 [0.954,1.032]	0.000	0	–	–	44.36 [38.84,45.57]
	HSM-gated (Dexter)	**1.075** [1.033,1.131]	**0.000**	**0**	**0.680**	N/A (0 fires)	45.24 [42.25,45.74]
	HSM-gated (`index_adv.`)	**1.073** [1.046,1.084]	**0.000**	**0**	**0.664**	N/A (0 fires)	44.90 [43.99,45.60]
	Periodic $K = 2$ (Dexter)	2.229 [2.123,2.283]	1.187 [1.10,1.21]	10	–	0.100	45.45 [42.61,46.22]
	Periodic $K = 2$ (`index_adv.`)	1.915 [1.774,2.246]	1.203 [1.11,1.52]	10	–	0.100	32.06 [31.25,32.78]
	Always-on (Dexter)	3.354 [3.305,3.438]	2.298 [2.26,2.35]	21	–	0.095	45.93 [45.53,47.52]
	Always-on (`index_adv.`)	3.194 [2.942,3.386]	2.388 [2.25,2.65]	21	–	0.095	43.60 [40.13,45.53]
Mid-Q: $Q = 700$ ($Q_{\min} < Q < Q_1$), $\theta^* = 0.826$, 10 blocks							
	baseline	18.47 [18.30,19.14]	0.000	0	–	–	42.69 [42.54,43.42]
	HSM-gated (Dexter)	**20.58** [20.38,20.69]	**0.159** [0.13,0.19]	**1**	**0.665**	**1.000** [1.000,1.000]	**42.59** [42.38,42.76]
	HSM-gated (`index_adv.`)	**14.65** [14.57,14.94]	**0.504** [0.50,0.51]	**1**	**0.711**	**1.000** [1.000,1.000]	**28.80** [28.54,29.24]
	Periodic $K = 2$ (Dexter)	41.73 [41.38,42.42]	21.97 [21.83,22.43]	209	–	0.010	43.18 [43.08,43.74]
	Periodic $K = 2$ (`index_adv.`)	31.28 [31.18,34.43]	19.10 [18.89,19.33]	209	–	0.010	29.55 [29.18,46.82]
	Always-on (Dexter)	61.43 [60.98,62.50]	40.80 [40.31,41.57]	420	–	0.005	44.26 [44.10,44.68]
	Always-on (`index_adv.`)	50.64 [50.36,51.43]	38.15 [37.87,38.42]	420	–	0.005	29.80 [29.52,31.59]
Large-Q: $Q = 3000$ ($Q \gg Q_{\min}$), $\theta^* = 0.959$ (theory), $\theta_{\mathrm{eff}} = 0.875$ (clamped), 10 blocks; baseline $t_{\mathrm{wall}} = 78.5$ s							
	HSM-gated clamped (Dexter)	**87.5** [87.3,87.8]	**0.18** [0.16,0.19]	**1**	**0.660**	**1.000** [1.000,1.000]	**42.64** [42.55,42.81]
	HSM-gated clamped (`index_adv.`)	**61.4** [61.0,61.7]	**0.50** [0.47,0.52]	**1**	**0.734**	**1.000** [1.000,1.000]	**28.67** [28.45,28.91]
	HSM-gated naive (Dexter)	145.5 [144.9,165.2]	58.5 [57.9,59.2]	601	–	0.003	42.71 [42.49,42.95]
	HSM-gated naive (`index_adv.`)	36.6 [36.2,38.7]	4.66 [4.55,4.78]	601	–	0.003	26.41 [26.18,26.71]

5.6 A-CE: Cross-Engine Validation on MongoDB

To validate HSM's DBMS-agnostic claim, we deploy the *same* kernel (W_0, DWT+SAX+FastDTW) on MongoDB 7 with the document-store extractor. A 24-window trace across four collection-specific phases (drift at W7, W13, W19) uses the identical 10-block RCB protocol as PostgreSQL. **A-CE1 ($\theta = 0.75$):** DR = 1.271; 61.7 % fewer advisor calls than always-on; Youden $J = 0.705$ (recall = 1.000, specificity = 0.705).

Threshold Calibration. At $\theta = 0.75$ (PostgreSQL parity), MongoDB shows $J = 0.705$ because the polymorphic single-collection design yields $\bar{S}_V = 1.0$ (no volume variation), lowering within-phase HSM to ≈ 0.78 vs. PostgreSQL's ≈ 0.88. A post-hoc ROC sweep reveals $\theta^*_{\mathrm{Mongo}} = 0.55$ ($J^* = 1.000$) vs. $\theta^*_{\mathrm{PG}} = 0.775$ ($J^* = 0.965$).

A-CE2 ($\theta = 0.65$, engine-calibrated): Re-running the *identical* workload (same seeds) with $\theta = 0.65$ yields DR = 1.486, $J = 0.981$ (specificity 0.981, FP 4/210 vs. 62/210), and 85.8 % fewer advisor calls—a $1.161\times$ speedup over always-on (vs. $1.011\times$ at $\theta = 0.75$). This confirms that *the HSM kernel is engine-agnostic; only θ requires per-deployment calibration*, consistent with the ROC framework of Theorem 4.

Cross-engine theta mismatch (B3 receipt). When the economically optimal θ from one engine is applied to a second engine without recalibration, precision degrades sharply. B3-A: PostgreSQL at $\theta = 0.55$ (MongoDB-optimal) yields zero triggers over 10 blocks; B3-B: MongoDB at $\theta = 0.775$ (PostgreSQL-optimal) yields ≈ 10 triggers per block with precision 0.27. This validates Theorem 4's prediction that $\theta^*(N, Q)$ is workload- and engine-specific, requiring per-deployment calibration.

A-CE3: Per-window breakdown schema verified identical (16 columns) across engines. The ROC curves and Youden-J sweep are in Supplementary §XIV.

5.7 Summary of Experimental Results

Table 4 summarises all A1–A14 and A-CE outcomes with their corresponding theorem validations. All six theorems and four lemmas receive consistent empirical support on PostgreSQL; cross-engine validation on MongoDB (A-CE) confirms the DBMS-agnostic property.

TABLE 4
Summary of experimental findings (A1–A14, A-CE, B1–B5). V. = validated against the indicated theorem or lemma. All six theorems and four lemmas receive consistent empirical support on PostgreSQL; B-series validates noise-floor properties and kernel/weight robustness.

RQ	Thm	Key result	V.
A1	L1	4 axioms verified; 0 violations ($n = 280$)	✓
A2	L1	$S_V \in [0.35, 1.00]$; unity for equal sizes	✓
A3	L1	S_T within 0.988; cross 0.983	✓
A4	L1	S_A drops 63.2% at phase boundaries	✓
A5	L1	DR = 1.624 (95%CI [1.613,1.636]); ICC = 0.952; $p < 0.001$; $\|r\| = 0.999$	✓
A6	T2,T3	S_T (-11.1%), S_A (-7.2%) critical; speedup $8\times$	✓
A7	L3	OLS slope = 0.935; $R^2 = 0.997$	✓
A8	Gen.	SDSS 1.310; JOB 1.250; TPC-B 2.008; burst 1.093	✓
A9	Util.	0.25% overhead; 919 trans/s	✓
A10	Rob.	$\leq 13\%$ DR change at $\pm 60\%$ noise; $\leq 4.1\%$ at ± 0.05 weight	✓
A11	T3	Break-even $N_{\text{pts}} \leq 17{,}000$	✓
A12	T5	$\beta = 1.177$ ($R^2 = 0.998$); $\Theta(N \log N)$ rebuild scaling ($N_{\text{lineitem}} \in [1.2\,\text{M}, 18\,\text{M}]$, SF 0.2–3.0). Kernel-cost micro-benchmark over $N_{\text{pts}} \in [100, 30{,}000]$ gives $\beta = 1.094$ ($R^2 = 0.9999$).	✓
A13	T4,T5	HSM-gated $2.92 \pm 0.18\times$ vs. always-on, $1.43 \pm 0.04\times$ vs. periodic; $80.8\%/42.5\%$ fewer calls; $61.9 \pm 1.2\%$ no-advisor retention; scale-invariant CV $< 7\%$ across SF $\in \{0.2, 1.0, 3.0\}$	✓
A13b	T4	$z \geq 3.40\sigma$ across scale; FAP$< 1\%$; $d' > 12$; Det. Prob = 100%	✓
A13c	T5	Bound $1/(1 - \hat{p}_{\text{stable}})$ realised on TPC-H ($5.22\times$, $80.8\%/42.5\%$), burst ($5.20\times$, $80.8\%/44.4\%$), SDSS ($4.01\times$, $75.1\%/25.2\%$), JOB ($20.0\times$, $95.0\%/85.7\%$); OLTP correctly diagnosed as high-drift ($\hat{p}_{\text{stable}} \to 0$, bound self-deactivates)	✓
A14	T4,T5	Youden $J \geq 0.9626$ across all SF; best at SF = 0.2 ($J = 0.9684$); 95 % certificate $[5.72\times, 23.45\times]$ at $\pi_w = 0.95$ (SF = 3.0 worst)	✓
A17	T6	Optimal N_{pts} from convex cost model; empirical min at $N_{\text{pts}} = 50$ (TPC-H SF 0.2, SF 3.0); $N_{\text{pts}} = 100$ (JOB)	✓
A-CE1	L1,T2	MongoDB DR = 1.271 ($\theta = 0.75$); $J = 0.705$	✓
A-CE2	T4,T5	$\theta = 0.65$: DR = 1.486; $J = 0.981$; 85.8 % fewer calls	✓
A-CE3	Repro.	Breakdown CSV schema: 16 columns, identical across engines	✓
B1	T3	Noise floor $S_{\text{floor}} \approx 0.875$; clamped $\theta = 0.875$ yields 1 call/block, P=1.000; naive $\theta = 0.959$ fires 601 times, P=0.003	✓
B2	T2	Kernel ablation: angular $J^* = 0.667$ (AUC=0.833), cosine $J^* = 0.667$ (AUC=0.833), euclidean $J^* = 1.0$ (AUC=1.0)	✓
B3	T3	Cross-engine mismatch: PG at $\theta = 0.55$ (Mongo-opt) yields 0 triggers; Mongo at $\theta = 0.775$ (PG-opt) yields $\approx$10 triggers, P=0.27	✓
B4	T2	Amplitude invariance: S_R, S_T constant across $k \in \{0.5, 1, 2, 5, 10\}$; floor ranges 0.695–0.875	✓
B5	T2	Robustness: θ^* stable 0.882–0.920 (CV$<1\%$) across noise/weight perturbations; AUC degrades at $\delta \geq 0.075$	✓

✓ = validated.

5.8 Discussion

Theory–Experiment Alignment. All theorems and lemmas receive consistent empirical support. Scale-invariant DR (CV = 0.1%) confirms $\theta = 0.75$ as robust. Inter-domain DR consistency (Supplementary, Dimensional-Utility Table) supports Theorem 2; weight sensitivity (A10b) confirms $\mathbf{w}_0$ lies in a broad flat optimum ($\leq 5.5\%$ DR change).

Threats to Validity. (i) Phase leakage (N5) most influential ($\pm 13\%$, A10). (ii) Three external validations (SDSS, JOB, pgbench) address TPC-H bias. (iii) OLTP may require weight recalibration; MongoDB (A-CE) tests data-model generality. (iv) J→P/P→C boundaries (HSM = 0.77–0.83) reflect genuine query-mix similarity.

Boundary Conditions. *Gradual drift* may keep $S_{\text{HSM}} > \theta^*$ as indexes degrade; operators should reduce N_{pts} or adopt a sliding-window variant. *Concurrent phase changes* can cancel in the composite score; per-dimension inspection provides diagnostic resolution.

Baseline Comparison. On a single workload, the best single dimension can match HSM (Table 5): S_A alone reaches $J = 0.993$ on TPC-H, S_T alone reaches $J = 0.832$ on SDSS. No single dimension is consistently strong, however: S_A drops to 0.227 on SDSS and S_T drops to 0.845 on TPC-H. The composite trades peak per-workload J for cross-workload consistency ($J \geq 0.58$ on SDSS, $J \geq 0.97$ on TPC-H), the formal metric properties of Lemma 1, the dimension-wise diagnosis of Algorithm 2, and the closed-form threshold of Theorems 3–5. Operators with known workload characteristics may further improve J by tuning $\mathbf{w}$ toward the dominant dimension.

Scope vs. ML-Based Advisors. BALANCE [5] and Indexer++ [7] solve the complementary *what*-to-build problem; a throughput comparison would conflate advisor quality with gate quality, and public implementations are unavailable.

TABLE 5
Baseline comparison: Youden J at optimal θ^* (all-pairs protocol). Bold: best single dimension per workload. No single dimension achieves $J > 0.85$ on all workloads; only the composite HSM achieves consistent cross-workload detection across all four benchmarks. Kernel-form rows (last block, marked †) are evaluated on the burst-mid-Q subset (kernel-ablation figure in the Supplementary, A13 Decision-Layer Cost; B2 receipt) and quantify pattern geometry rather than dimension utility.

Method	TPC-H SF 0.2	SDSS	MongoDB
S_R only	0.979	0.416	1.000
S_V only	0.000	0.001	0.000
S_T only	0.849	**0.832**	**1.000**
S_A only	**0.993**	0.227	0.908
S_P only	0.022	0.086	0.985
$S_T + S_A$ (2-dim)	0.962	0.652	1.000
Uniform 5-dim	0.969	0.575	1.000
Angular kernel†	0.667	–	–
Cosine kernel†	0.667	–	–
Euclidean dist.†	1.000	–	–
HSM ($\mathbf{w}_0$)	**0.971**	**0.576**	**1.000**

A12/A14: Scale Sensitivity and Detector Quality (Theorems 4, 5). OLS over $N_{\text{lineitem}} \in [1.2, 18]\text{M}$ (SF 0.2–3.0) yields $\beta = 1.177$ ($R^2 = 0.998$) for rebuild cost $T_A(N)$ and $\beta = 1.094$ ($R^2 = 0.9999$) for kernel cost $T_{\text{HSM}}(N_{\text{pts}})$ over $N_{\text{pts}} \in [100, 30{,}000]$; both are consistent with the Lemma 3 bounds (the finite-range super-linear residual absorbs Fast-DTW path length and memory-hierarchy effects without a genuine $\log N_{\text{pts}}$ factor), and the gap between them sources the $O(N_{\text{pts}}/(N \log N))$ amortisation of Theorem 5. Youden $J \geq 0.9626$ at every scale ($J^* = 0.9684$ at SF = 0.2, 0.9626 at SF = 3.0, both at $\theta^* = 0.775$). Theorem 5 gives a 95 % certificate $[5.72\times, 23.45\times]$ at $\pi_w = 0.95$ (SF = 3.0 inputs), central $16.1\times$,

rising to $41.9\times$ at $\pi_w = 0.99$. Full per-SF tables, Hoeffding bands, and ROC sweeps: Supplementary §XII.

6 Conclusion

HSM is a five-dimensional, training-free similarity measure for proactive index management, with a formal core (six theorems, four lemmas) covering metric soundness, $\Theta(N_{\text{pts}})$ complexity, an economic gating threshold, stratified-Hoeffding detector bounds, a speedup certificate $\rightarrow 1/(1-p_{\text{stable}})$, and a convex-optimal window size. Experiments confirm each theorem (Table 4): scale-invariant DR across SF 0.2–3.0, $< 0.3\%$ overhead, and a Theorem-5 95% deployment certificate $[5.72\times, 23.45\times]$ at $\pi_w = 0.95$ on the worst-case SF = 3.0 inputs.

Limitations. (i) J→P and P→C boundaries (HSM = 0.77–0.83) are not flagged because the underlying query mixes are genuinely similar. (ii) The fixed weights $\mathbf{w}_0$ deliberately target cross-workload consistency rather than peak per-workload Youden J: on workloads where a single dimension carries the phase signal (e.g. S_T on SDSS), a $\mathbf{w}_0$-tuned single-dimension detector matches HSM on peak J (Table 5) but forfeits the composite metric properties of Lemma 1 (triangle inequality required by the Theorem 3 break-even analysis) and the dimension-wise diagnosis of Algorithm 2. HSM therefore retains the five-dimensional composite as the default; automated per-workload weight calibration that preserves P1–P4 is left to future work. (iii) End-to-end throughput is reported only for SF $\in \{0.2, 1.0, 3.0\}$ (up to ≈ 18 M rows); the kernel-measurement cost is tight-$\Theta(N_{\text{pts}})$ (Lemma 3) and extrapolates, but end-to-end runs at SF ≥ 10 are dominated by buffer-pool, autovacuum, and unindexed-join effects outside the kernel and are deferred to future work. (iv) Theorem 5's operational speedup is workload-specific by construction and asymptotes to $1\times$ on intrinsically high-drift workloads such as pgbench OLTP ($p_{\text{stable}} \rightarrow 0$); the bound remains exact, and the gate's TPR = 1.00 at default θ correctly reports the diagnosis. *Future work:* live advisor integration; multi-tenant cloud; automated θ calibration; extended NoSQL validation; SF ≥ 10 end-to-end profiling. Complete proofs are in Supplementary §I–X.

Code and Data Availability

A complete replication package [51]—source code, configuration, scripts, and instructions to reproduce all reported results—is publicly available at https://github.com/stevezhai139/HSM_gated_5 (BSD-3-Clause). The package includes a five-minute reviewer quickstart (CPU-only path, no database required), the metric-axiom unit tests for Lemma 1, the HSM kernel and θ^* implementation, per-workload runners (TPC-H, SDSS, JOB/IMDB, pgbench, burst), the end-to-end Dexter and Supabase `index_advisor` harnesses, and the shipped CSVs that back every figure and table in the paper. Step-by-step reproduction of the PostgreSQL 16 and MongoDB 7 end-to-end runs is documented in `REPRODUCE.md`. A versioned snapshot will be archived at Zenodo upon acceptance.

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